Section 2b Mechanical installation (Part 1)

Contents

1. CHECK OF THE MAGNET SHOCK-INDICATORS

Shock indicators must be checked before unloading the magnet from the truck in order to define responsibilities in case of damage to the magnet.

If one or more shock indicators are tripped, the magnet did withstand higher mechanical forces than it was designed for. This means there is a risk of internal (invisible) magnet damage.

NOTE

- *When you accept a magnet, which has a shock-indicator, tripped: make a note on the waybill to be countersigned by the transporter.*
- *Only shock indicators mounted on the magnet, not on the magnet crate, are valid for this procedure*

Remove all shock indicators from the magnet.

If cold spots (condensation) are visible on the surface (not on the turret) call the Helpdesk MR before moving the magnet into the hospital.

It is important to check a.s.a.p. whether the magnet is damaged or not. Access to the room should be maintained in the event the magnet has to be replaced. Following information has to be collected and sent to the Helpdesk MR in Best

a. Which shock-indicator(s) (position / direction / 10 and/or 15 G) are tripped?
Remark: If 5G indicators are tripped, no actions are required. Only make a note on the waybill.

b. Does the cryostat have any visible damage? If so, make pictures including pictures of the packing.

c. Copy of the waybill, including the notes you have made on the waybill.

d. 12 planes bare plot, including a description of the iron near the magnet.

e. Helium boil-off figures.

These measurements will be compared with the data before shipment in order to advise you about the actions to be taken.

Possible actions are:

- The magnet performance is still within specification. Continue with the installation.
- The magnet can be locally repaired.
- The magnet has to be replaced.

Ad a:

In the factory twelve shock indicators are mounted on the magnet.

There are two kind of shock-indicators:

1. Type 1 indicator is built up of four metal balls and two springs. When not tripped, each spring presses two balls (one at each side) against the housing. One spring is mounted in horizontal direction and one in vertical direction.

   As soon as the shock indicator (and the magnet) is accelerated faster than specified in a certain direction, the balls fall out of their position. In this situation the shock indicator has been tripped.

2. Type 2 indicator has an arrow that turns blue in one of the two directions when tripped.

   No actions are required if only type 1 shock-indicators are tripped

Ad e:

"Boil-off" is the evaporation of liquid helium inside the He-vessel.

Two methods can be used to measure the boil off:

1. Make two readings with an interval of several days. Divide the difference of these two readings by the time (in hours) between these two readings.

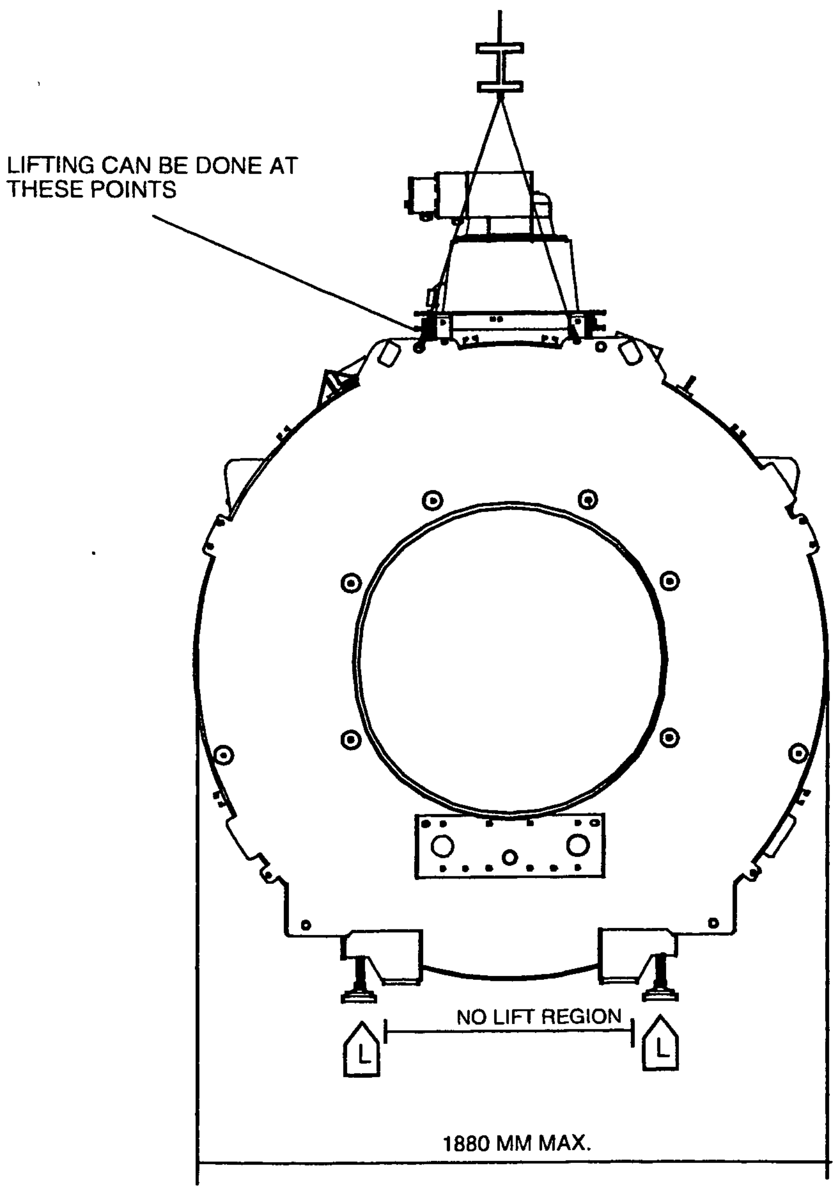
$$\text{Boil off} = [l/h] \qquad \frac{(\text{Reading at } t1\ [l] - t2\ [l])}{t2 - t1\ [\text{hours}]}$$

2. Connect a helium calibrated flowmeter to the gas exhaust unit, using the necessary reducing pieces and a hose (inner diameter > 5 mm, length > 5 m). Do a leak test with snoop (supplied with the system) to make sure all the gas is coming trough the flowmeter. Let the pressure stabilize for 12 hours, do not change the magnetic field and avoid cryogenic actions. After 12 hours or longer measure the gas-flow Multiply the gas-flow (litre/minute) by 0.090 to get the liquid helium consumption in litre/hour. The most accurate way is to take the mean value over a period of one week to take into account variation of boil off due to changes in atmospheric pressure. (2 readouts every day). The boil-off figures (< 1.0 l/h with refrigerator system switched off). are specified as an average over 5 days with gradients NOT pulsing.

2. TRANSPORTING THE MAGNET

The magnet may only be lifted using the lifting eyes located on top of the magnet or by lifting it at the designated areas on the bottom of the magnet. Lifting cable angle and points must be in accordance with the markings For further information refer to the "Shipping, handling & storage" document supplied with the magnet.

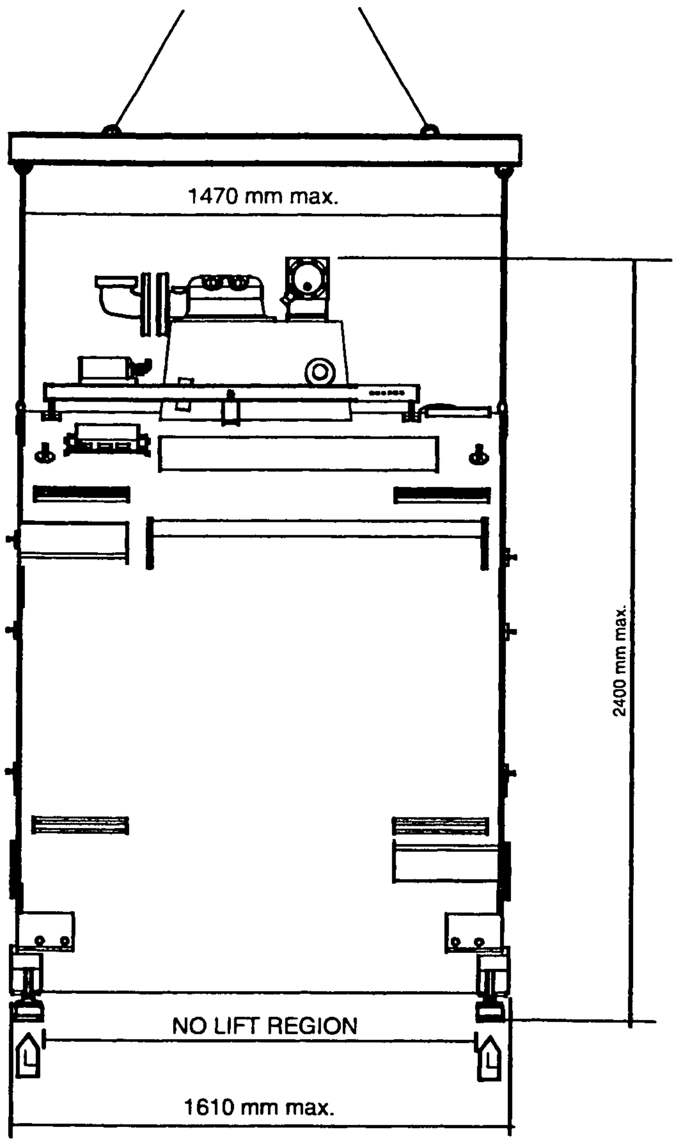
Figure 1: Magnet transport



MAGNET WEIGHT

1 5T <3400 kg
1 0 T <3200 kg
0.5 T <3000 kg

Figure 2 – Magnet transport



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CAUTION

The "magnet transport set T5" may not be used for NT magnets. The extensions of this set are to weak. Instead use "magnet transport set T5/NT/Intera" (see Section 1 Service Tools).

Cover the floor with wooden or metal plates to prevent the floor from being damaged

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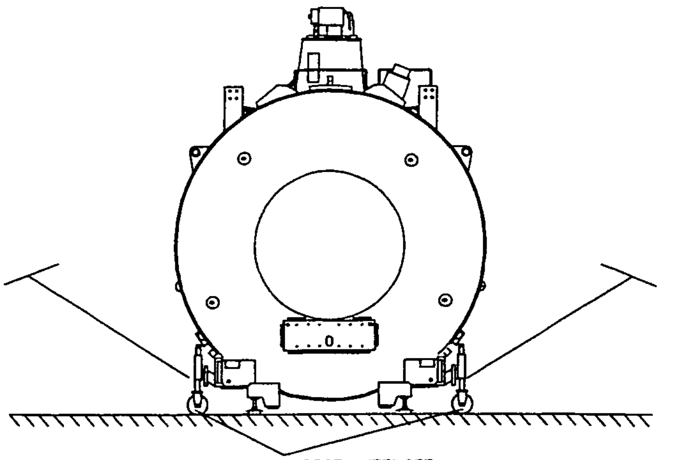
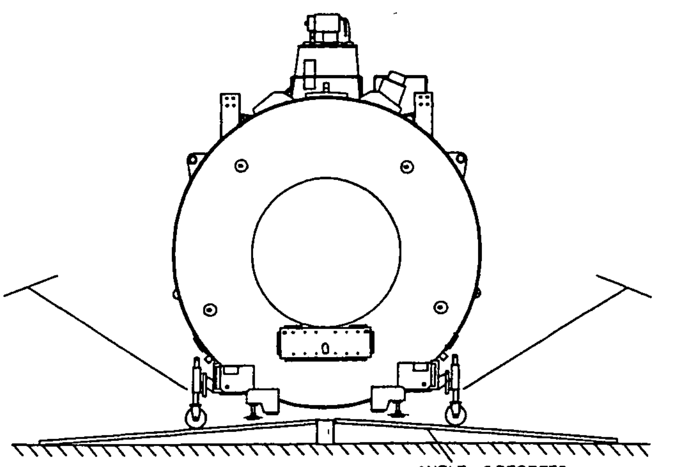
A set of 4 caster wheels can be mounted to the magnet for transporting it into the RF room.

Figure 3- Wheel set for magnet transport



If a ramp is used to transport the magnet over a threshold, the angle of the ramp may not be more than 5 degrees. Otherwise the magnet will touch the ramp The magnet can be pushed over the ramp with four or five people. Make sure that the turret of the magnet does not touch the RF cage.

Figure 4- Magnet transport over a ramp



3. TRANSPORT OTHER SYSTEM PARTS

For transporting the gradient coil combination, the forks of the pallet truck must have a minimal length of 150 cm.

When cabinets are damaged, make a note on the waybill, to be countersigned by the transporter. Contact your local Customer Support manager to discuss further steps. If possible make photos.

3.1. CHECK OF THE COPLEY GRADIENT AMPLIFIER SHOCK-INDICATORS

Shock indicators must be checked after unpacking in order to define responsibilities in case of damage to the Copley gradient amplifier.

If one or more shock indicators are tripped, the amplifier did withstand higher mechanical forces than it was designed for. This means there is a risk of internal (invisible) damage inside the cabinet.

There are two kind of shock indicators used:

- One type of indicator has an arrow that turns blue in one or two directions when tripped. These Shock indicators can be seen from the rear side of the cabinet.
- The other shock indicators have a small horizontal tube, which turns red when tripped. They are on the top cover inside the Copley cabinet.

NOTE

When you accept a Copley gradient amplifier, which has a shock-indicator, tripped: make a note on the waybill, to be countersigned by the transporter. If possible make photos.

In case of tripped shock indicators the following information has to be collected and sent to the Helpdesk MR in Best:

- The system reference number
- Copley serial number (At the front upper right corner of the Copley).
- Which shock-indicator(s) (position / direction / 15 G and/or 25 G and/or 50 G) are tripped?
- Does the gradient amplifier cabinet has any visible damage? If so, make photos including photos of the packing.
- Copy of the waybill, including the notes you have made on the waybill.

If there is no visible damage, you can continue the installation.

4. POSITIONING THE MAGNET

Due to the fact that treated holes for the patient support are already present in the floor pads, it is needed to position the magnet in respect to these holes. For accurate positioning it is required to position the magnet using the supplied magnet-positioning jig.

The magnet must be placed on the four pads of directly on an epoxy floor if this has sufficient strength. No fixation to the floor is needed, unless required by local regulations.

In earthquake areas special precautions must be taken for mounting the magnet to the floor (for details refer to the Planning Reference Book). The mounting must be done in such a way, that the magnet remains insulated from the RF cage. If the mounting points are part of the RF- shielding, the construction must comply with the local regulations.

The distance between the rear magnet interface plate to the finished wall or, in case something is placed on that wall, to the protruding part must be at least 760 mm.

4.1. TOOLS

- Crowbar (length >= 1.5 metre)
- Magnet transport set
- Spirit level, length 500 mm, accuracy 0.1 mm/m
- Spirit level (3 m transparent water hose)
- Ruler
- 4 bolts M10x100 mm (only for NT patient support)
- Two open-ended spanners, size 30
- Digital Volt meter
- Magnet positioning Jig (for MT patient support only)

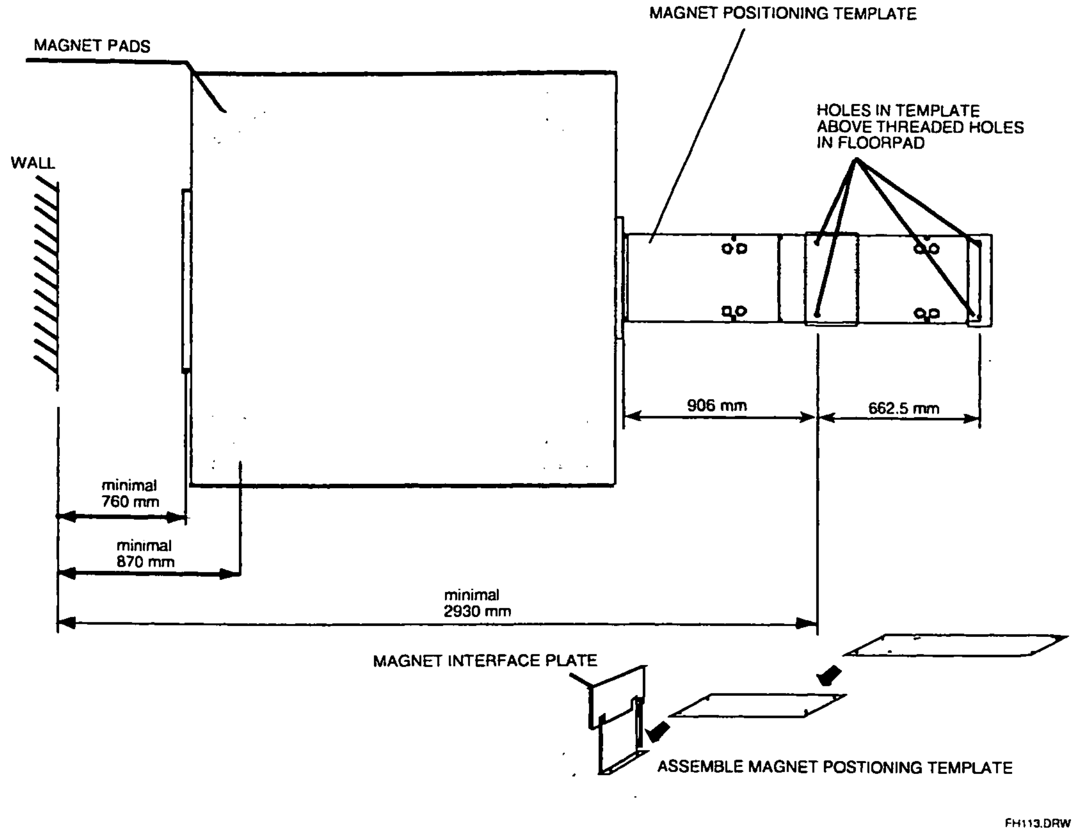
4.2. PROCEDURE

Check that the distance between the centre-point of the two rear magnet pads to the rear wall, or in case something is placed on the wall, to the protruding part is >= 870 mm. If no pads are present, measure the distance between the patient support fixation points (closest to the rear RF wall) to the rear wall, or in case something is placed on the wall, to the protruding part is >= 2930 mm.

1. Mount the four magnet feet underneath the magnet. The magnet feet are supplied with the magnet in a separate box.
2. Place the magnet in position using the magnet transport wheels.
3. Mount the magnet positioning jig to the magnet.
4. Position the magnet (burst disk towards the rear) so that the four holes in the magnet positioning jig are positioned above the threaded holes in the floor pads. Position the magnet as good as possible using the transport wheels. For exact positioning loosen the wheels and position the magnet using the crowbar.

Site assembled magnet system					
Intera 0.5T	Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master

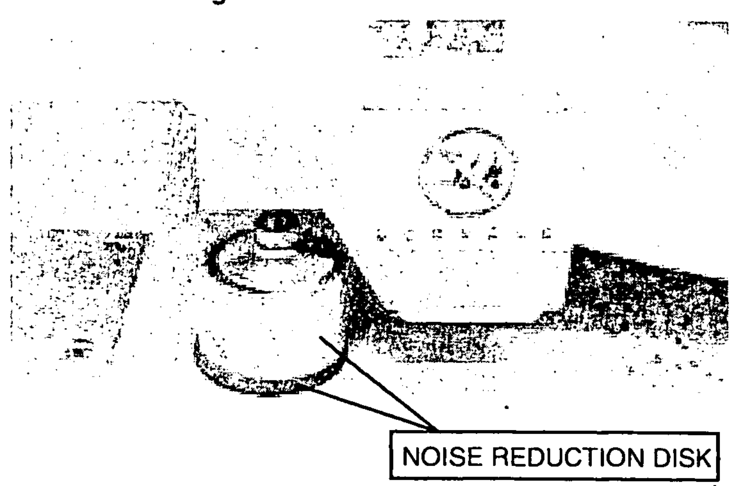
Figure 5 - Magnet position



After placing the noise reduction disks underneeth the magnet, the weight of the magnet must be spread equally between the four disks. If the magnet weight is spread between two disks too long (more than 1 hour), the disks might distort.

5. If the magnet is in the correct position, place one noise reduction disk underneath each foot.

Figure 6- Noise reduction disk

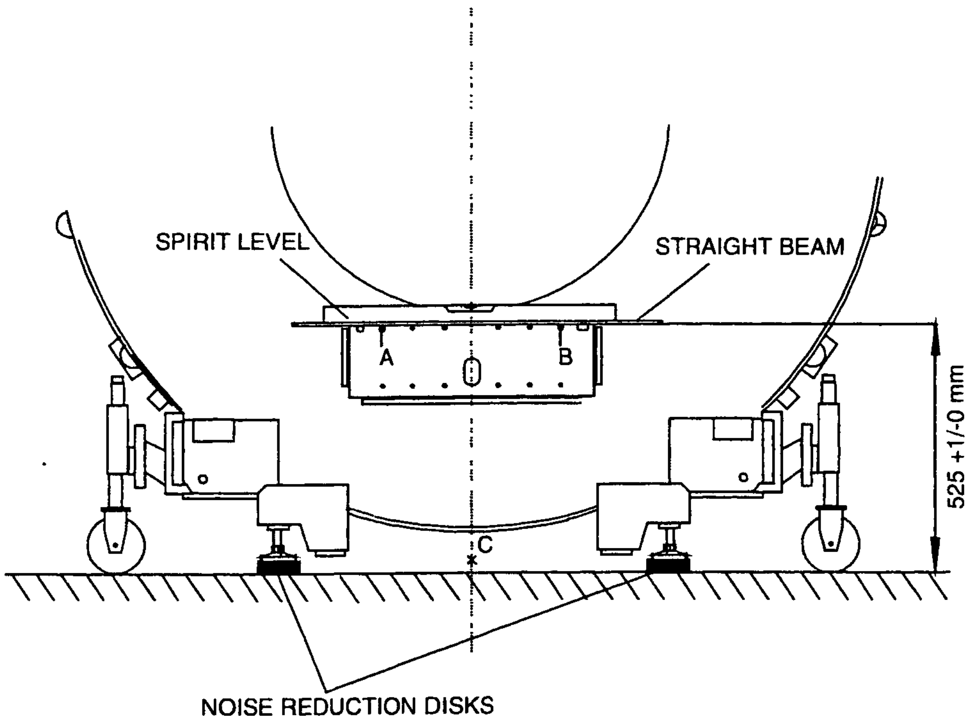


5. ADJUST THE HEIGHT OF THE MAGNET

Height adjustment and levelling can be done by using the four adjustable feet of the magnet Use the magnet transport wheels to lift the magnet.

1. Place a straight beam on top of the two bolts A and B at the front interface plate.

Figure 7- Height adjustment of the magnet



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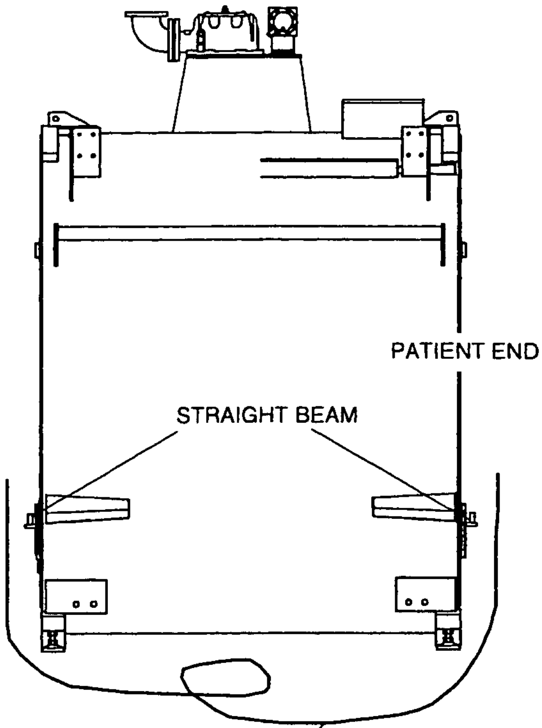
2. Adjust the height to: distance between the top of the bolts A, B and the finished floor: 525 + 1,- 0 mm (only at the front side).

6. LEVEL THE MAGNET

Figure 8 - Longitudinal levelling of the magnet



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1. Only at the front of the magnet, level the magnet perpendiculary (side ways) using a spirit level. Apply equal force to the nuts of all feet.
2. Place another straight beam on top of the bolts A and B (see Figure 7) at the rear interface plate. Level the magnet in the longitudinal direction, use a transparent hose filled with water.

   Measure at both ends of the straight beams. The centre of the rear beam must be levelled, within 1 mm, with respect to the straight beam at the front interface plate. Apply equal force to the nuts of the two rear feet.

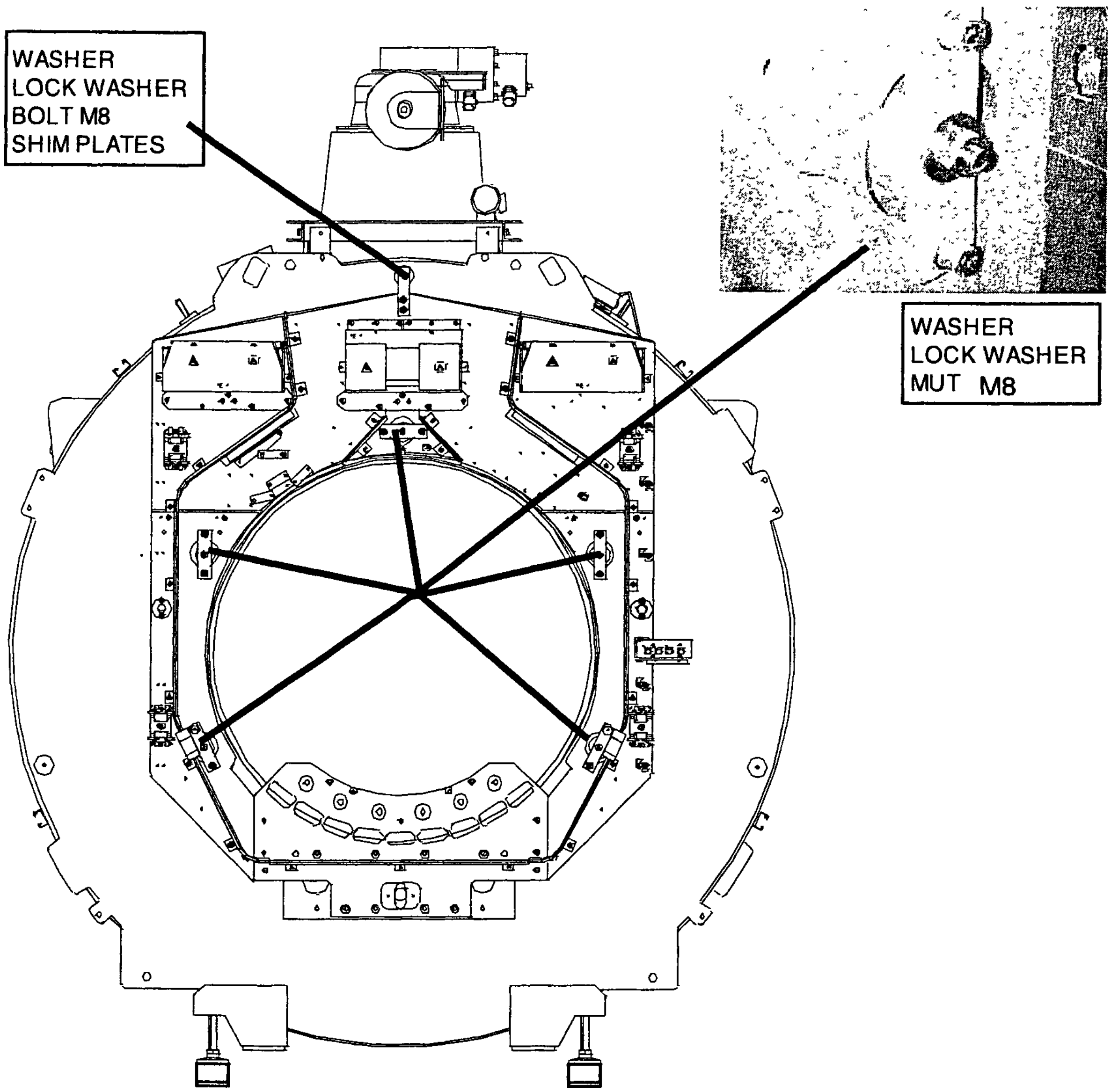
3. Lift the wheels, until the wheels are free from the floor. Check with the digital volt-meter the resistance between the magnet and the RF cage. R >= 200 Koml. Correct insulation, if not within limits. Write the measured value in the MRL.
4. Lock the adjustments made by fixing the nuts on top and bottom of the threaded rods of the magnet feet. Use two spanners of 30 mm.
5. Remove wheels.

7. MOUNTING THE RING-FRAME REAR

Install the rear ring-frame. The ring-frames are factory pre-mounted and adjusted. They must be installed before installation of the gradient carriers.

Mount the ring-frame at the inner five fixation points (Do not tighten the nuts).

Figure 9 - Mount the rear ringframe



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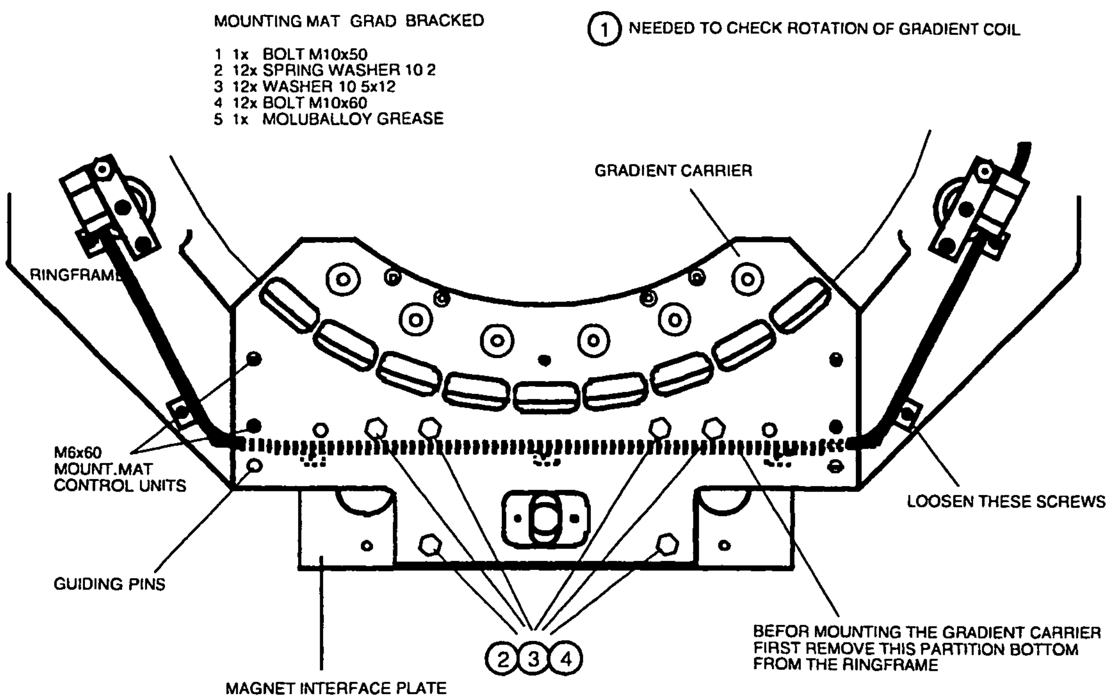
4. Mount the rear gradient carrier to the magnet. At the same time mount the ring-frame to the gradient carrier. Put some grease (Molykote) on the bolts.

Figure 10 - Mount gradient carrier



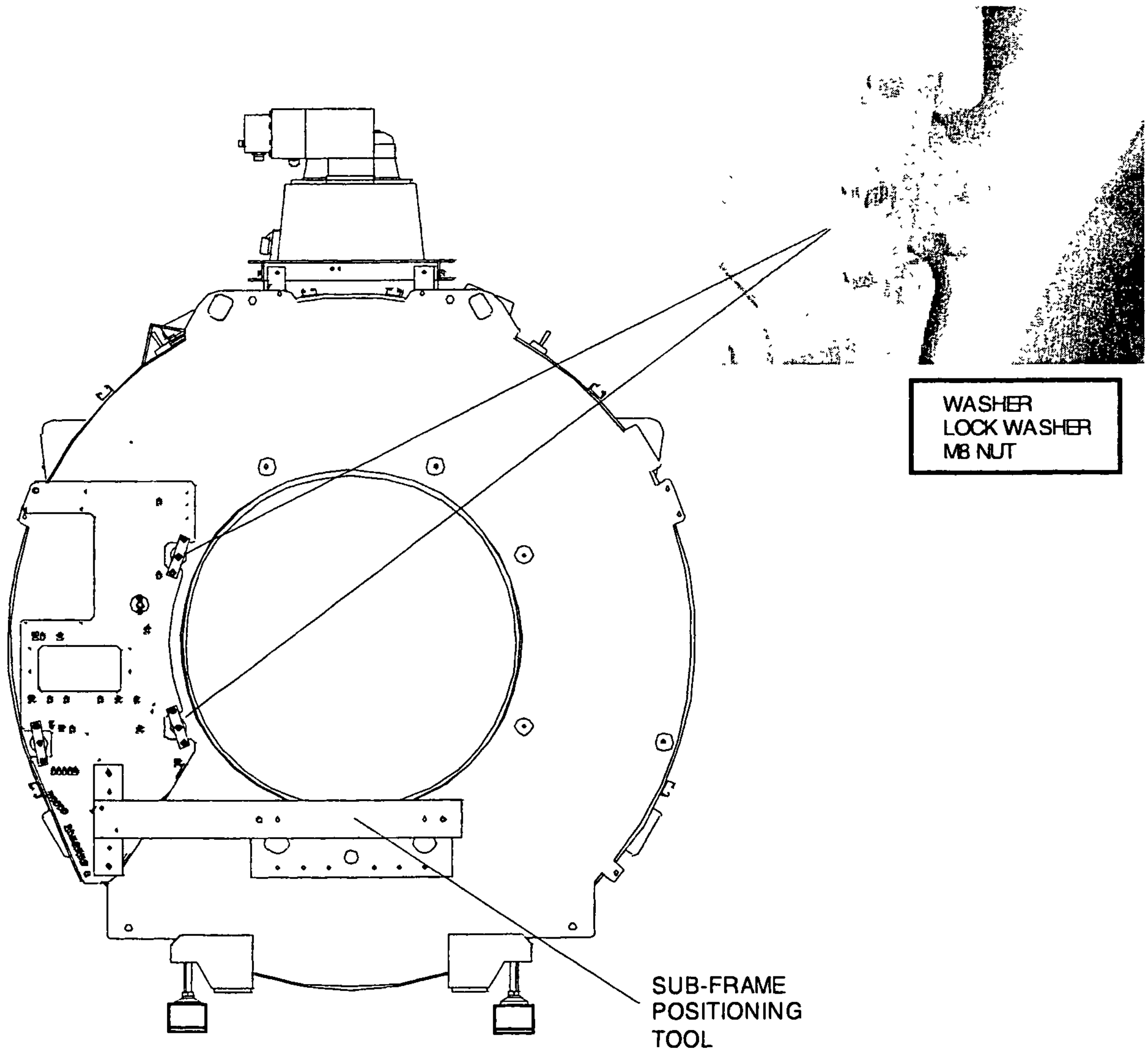
5. Tighten the ring-frame to the gradient carrier.
6. Tighten the ring-frame to the magnet at the five fixation points.
7. Shim-plates must be used at the upper fixation point.

8. MOUNT THE FRONT SUB-FRAMES

8.1. MOUNT THE LEFT SUB-FRAME

1. Mount the "sub-frame positioning tool"to the left sub-frame left.
2. Mount the tool together with the sub-frame to the magnet. Mount the tool to the magnet interface plate with two bolds M10. Two guiding are used for excact position.
3. Fix the left sub-frame only at the inside (See figure) and tighten these nuts.
4. Remove the "sub-frame positioning tool".

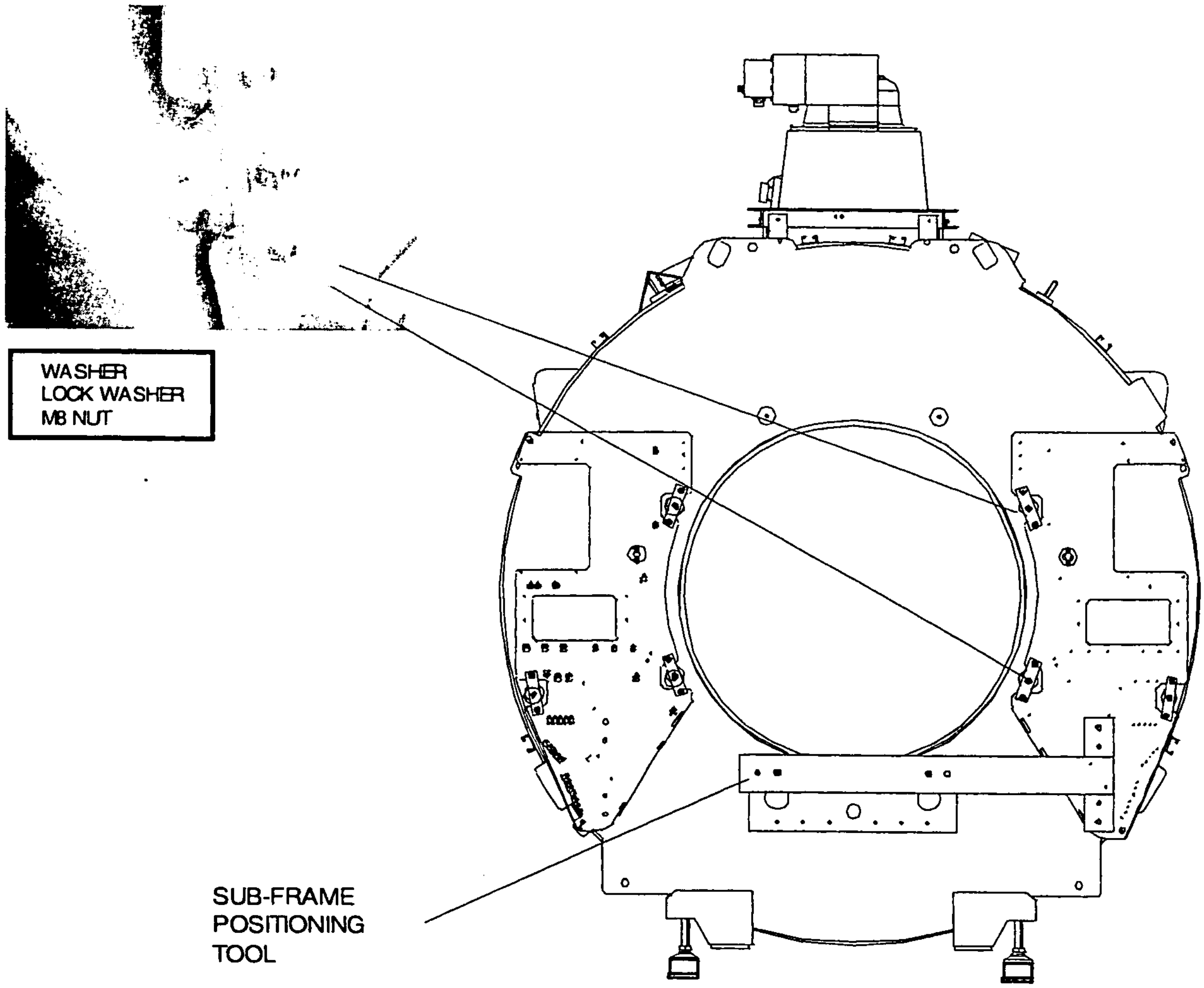
Figure 11 - Mount the left sub-frame



8.2. MOUNT THE RIGHT SUB-FRAME

1. Mount the "sub-frame positioning tool"to the sub-frame right.
2. Mount the tool together with the sub-frame to the magnet. Mount the tool to the magnet interface plate with two bolds M10. Two guiding are used for excact position.
3. Fix the left sub-frame only at the inside (See figure) and tighten these nuts.
4. Remove the "sub-frame positioning tool".

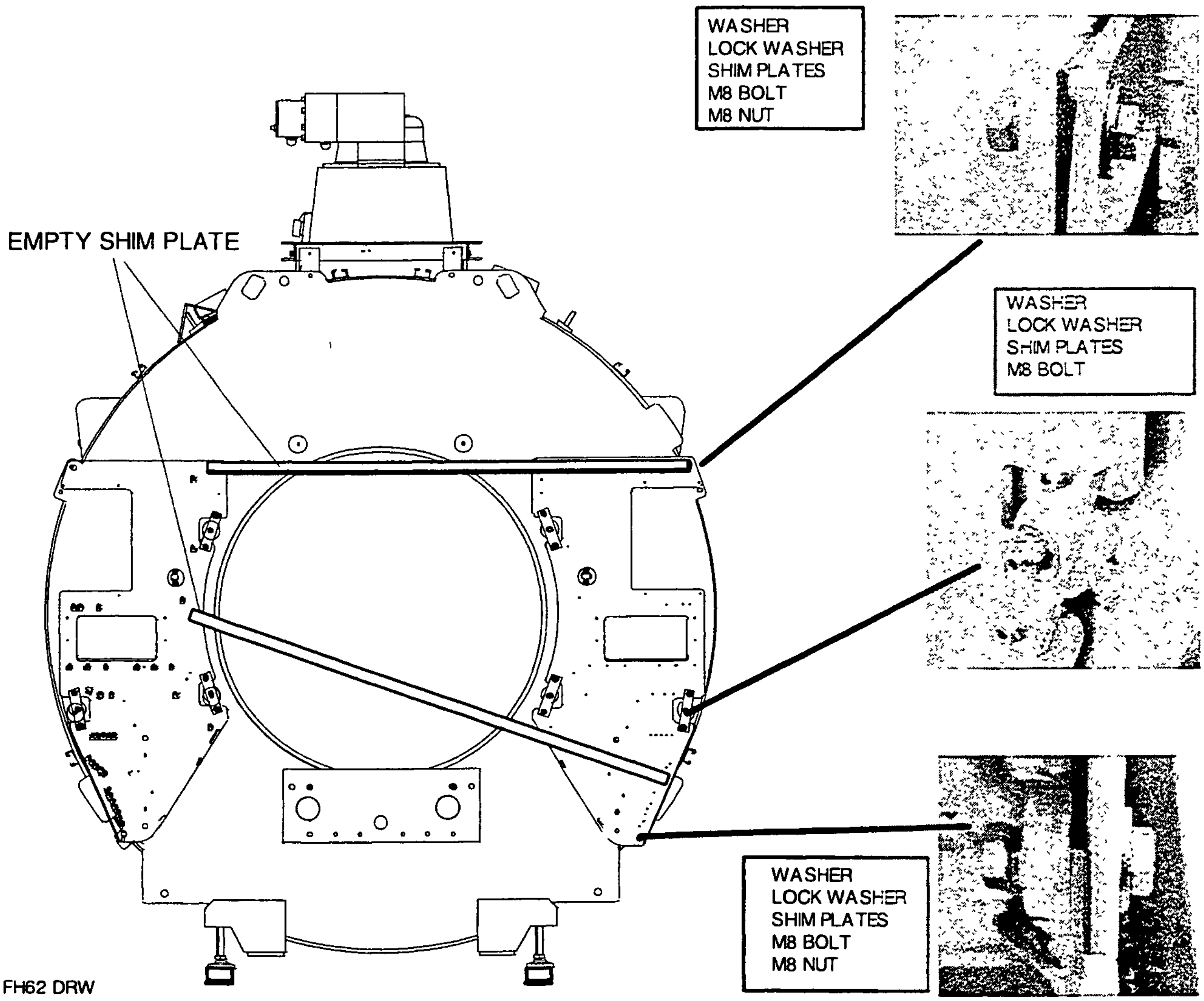
Figure 12 - Mount the left sub-frame



The inner points of the magnet for fixation of the sub-frames are acurate. The outher points however are not acurate. To mount the left and right sub-frames in one plane, use an empty shimstrip and hold it against the sub-frames (see figure).

Mount the outher fixation points (see figure) and add spacers untill the two sub-frames are in one plane.

Figure 13 - Mount the left sub-frame



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9. MOUNTING THE GRADIENT COIL

Using the gradient coil packing, the gradient coil can be mounted inside the magnet bore. The weight of the gradient coil is 670 kg.

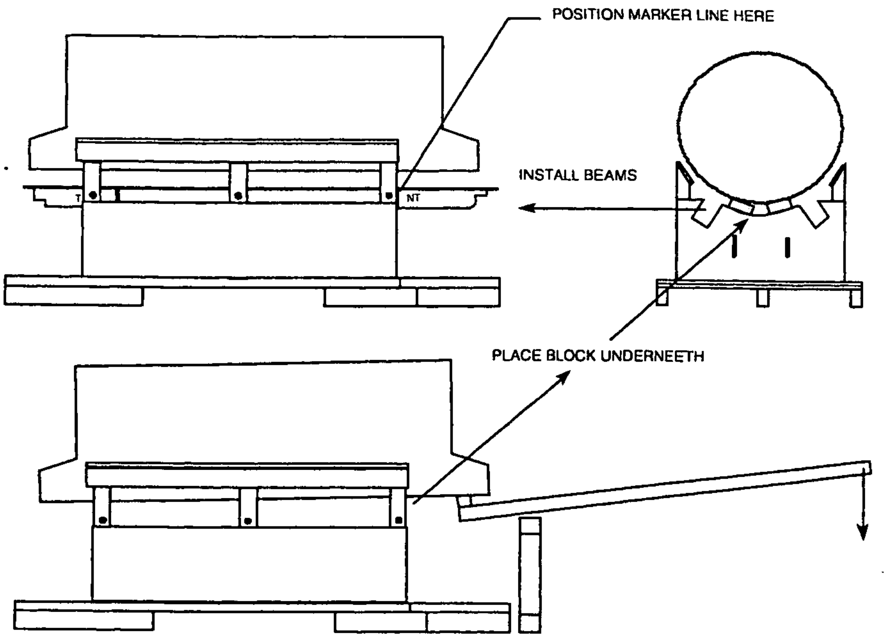
At this stage it is assumed that the rear gradient carrier with ring-frame has already been installed.

NOTE

To prevent mistakes during shimming, write the position numbers at the shim-rails to the front of the shim-rail and magnet. Use a black felt-marker.

1. Remove all shim rails from the magnet and clean the inner-bore with a vacuum cleaner.
2. Position the gradient coil with packing near the front of the magnet, using a pallet truck.
3. Open the packing.
4. Lift the gradient coil with the delivered beam and position a block of wood underneath it at rear and front side.

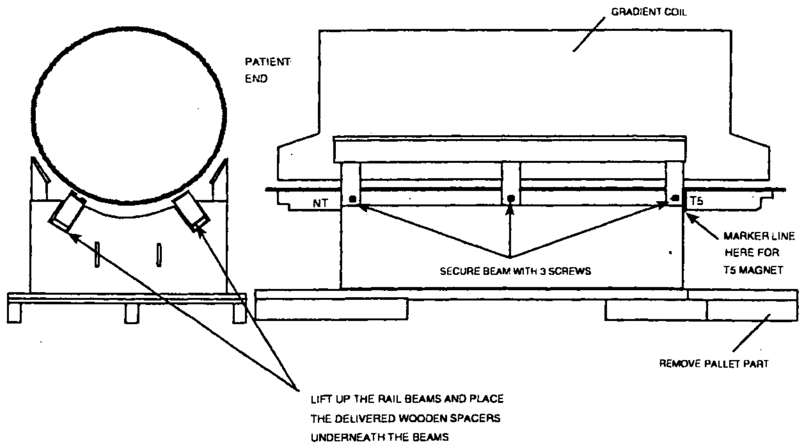
Figure 14- Lift gradient coil and install beams



5. Slide the two rail beams into the packing/cradle with the description NT towards the removable pallet part.
6. Shift the beams so that the marker lines on the beams are in line with the front of the packing/cradle.

7. Lift up the beams towards the gradient coil and insert a wooden spacer between the packing and the beam (on six locations).
8. To secure the beams, insert three screws from the side into each beam.

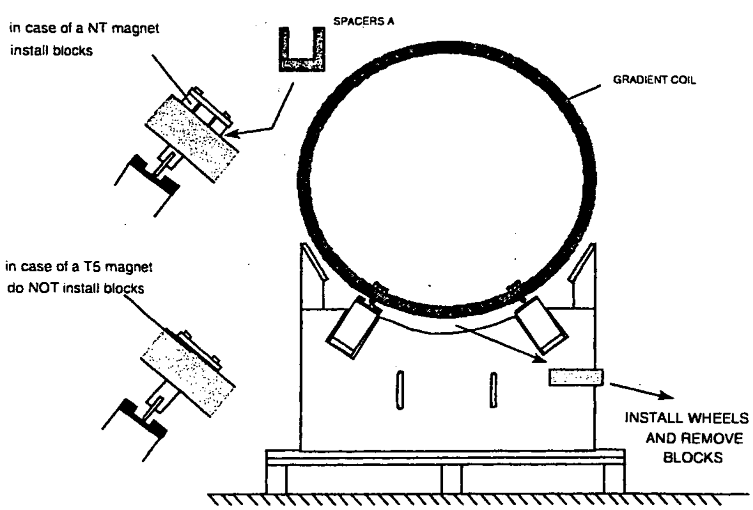
Figure 15 - Position beams and secure the beams



9. Mount the four small wheels into the gradient coil together with the spacer blocks.

Do not use the thin metal spacers A (see Figure 16) at this moment. If you need more space between the gradient coil and magnet bore at the bottom, you can loosen the screws that fix the wheels. In that case insert the spacers A between the wheel and the gradient coil. After inserting these spacers tighten the screws again.

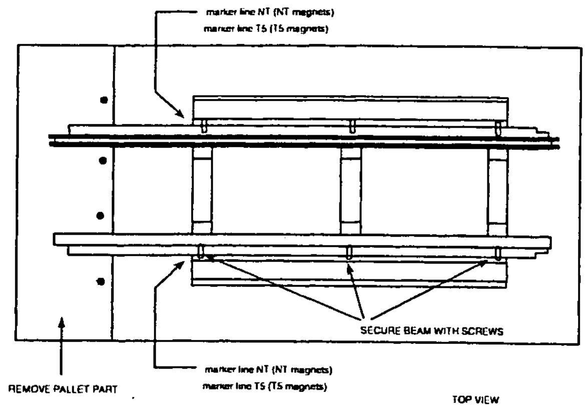
Figure 16 - Install wheels



10. Remove the wooden blocks underneath the gradient coil so the gradient coil stands free on the beams.

11. Detach the front part from the pallet.

Figure 17- Remove pallet front part

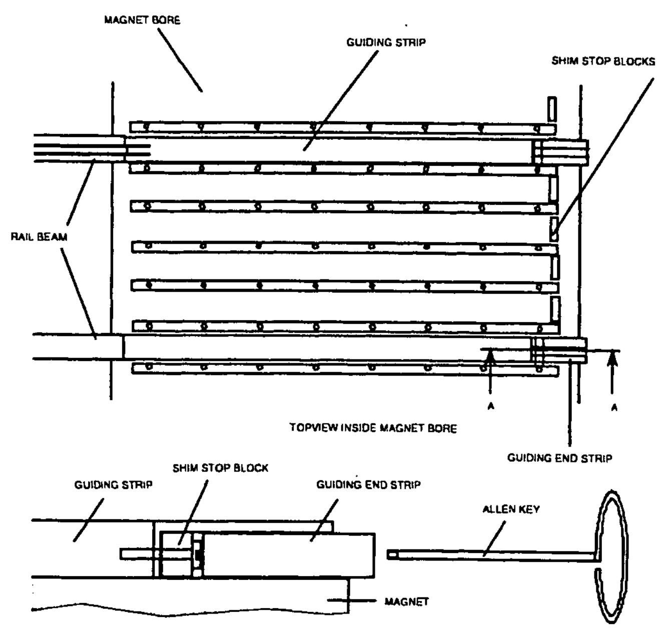


CAUTION

The gradient coil may not be installed into the magnet without protecting the magnet bore. Therefore guiding strips must be installed first. They are delivered with the gradient coil.

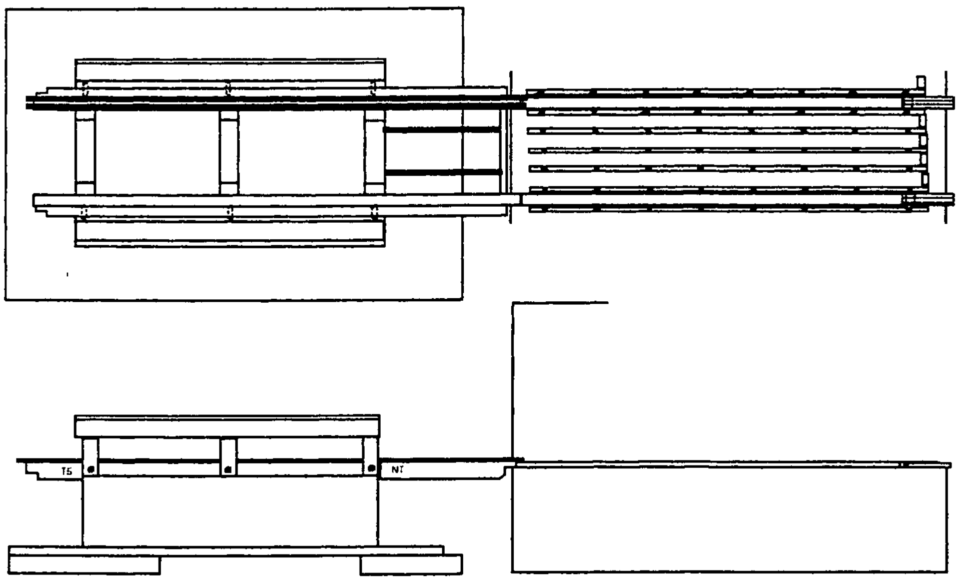
12. Position the two guiding strips and guiding end strips inside the magnet bore (position 16 and 22). Fix the guiding end strips to the guiding strips using the allen key of 8 mm. Position the guiding end strips from the back underneath the guiding strips.

Figure 18 - Install guiding strips



13. Lift the pallet with the pallet truck and position it in front of the magnet. The two rails must be placed in line with the guiding strips.

Figure 19 - Align the packing to the magnet



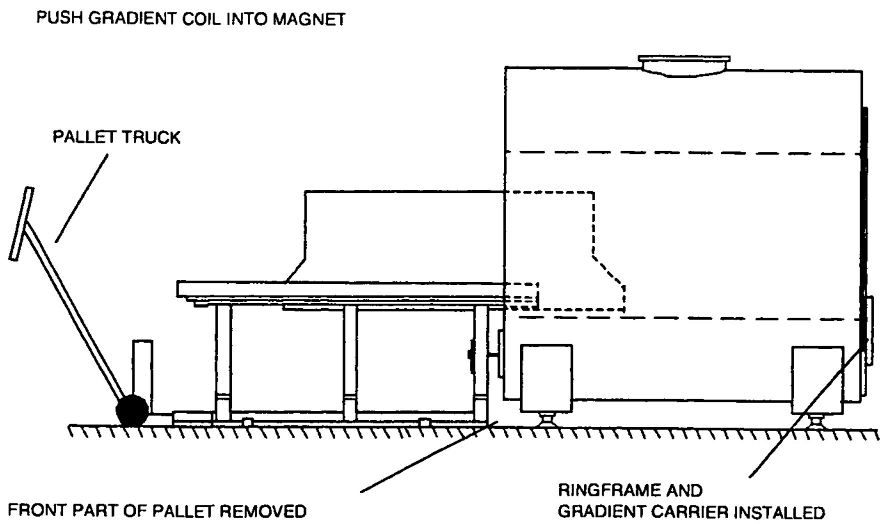
14. Position the rail beams so that they touch the guiding strips. Lower the packing/cradle so that the rails touch the guiding strips.
15. Mount the two threaded rods and bolts to the magnet interface plate and mount them to the packing so that the packing can not move backwards when inserting the gradient coil into the magnet.
16. Place the 4 delivered wedges underneath the packing so that it can not accidentally drop.

CAUTION

Do not push the gradient coil with excessive force against the rear gradient carrier. The magnet interface plate might be damaged.

17. Slide the gradient coil carefully into the magnet bore and push it carefully against the rear gradient carrier.

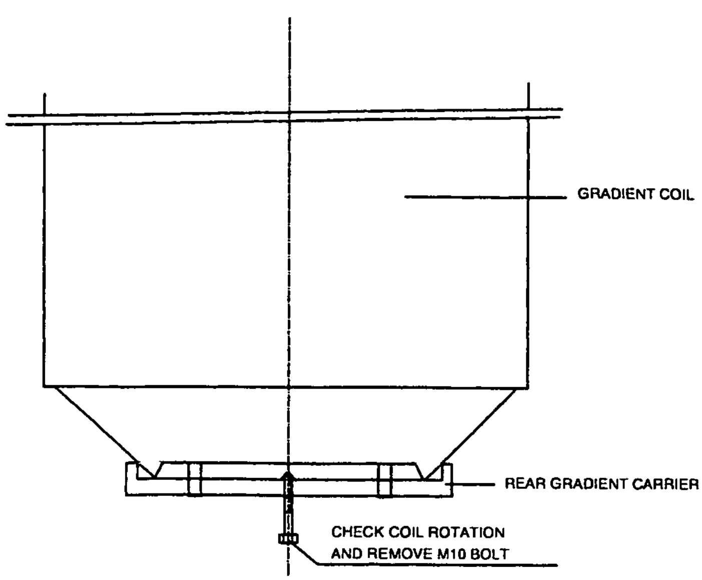
Figure 20 - Slide gradient coil into the magnet



18. Install the front gradient carrier on the front magnet interface plate.
19. Push the gradient coil to the front gradient carrier.
20. Lower the gradient coil by removing the four wheels. Remove also the guiding rails and the Guiding end strips.
21. Screw a M10 bolt into the rear gradient carrier to check the coil rotation. **Do not tighten this bolt** and **do not use it to push the gradient coil**. This causes damage to the magnet interface plate.

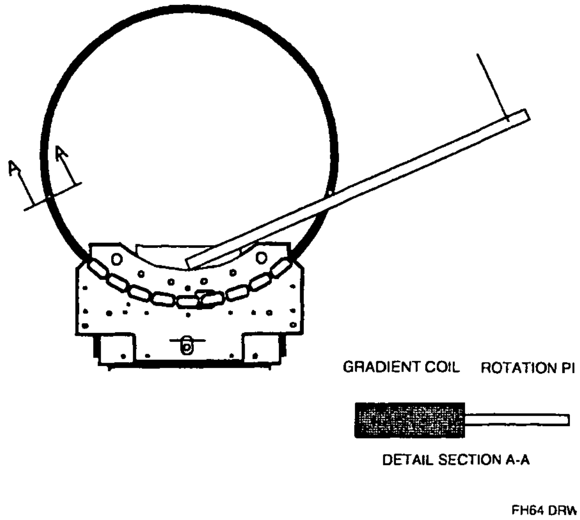
The bolt must align with the cutout in the gradient coil. If it does not, rotate it accordingly.

Figure 21 - Check coil rotation



23. If needed rotate the gradient coil using two rotating pins. Install them into the front of the gradient coil. Rotate the coil using a wooden beam.

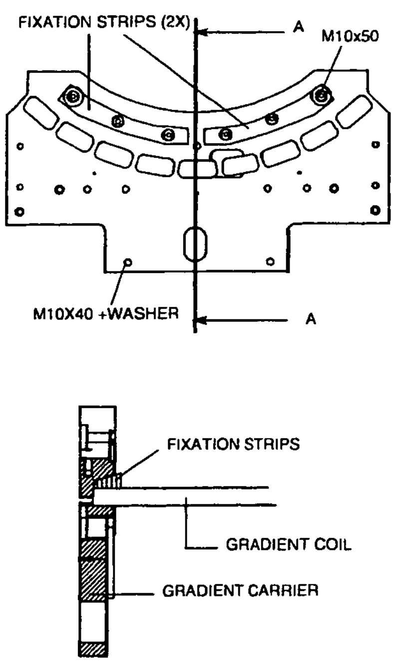
Figure 22 - Rotating the gradient coil



After checking/rotating remove the M10 bolt.

24. Fixation of the coil to the gradient carrier must be done with the four fixing strips. **Start with fixing the front side.** The four fixing strips are made of aluminium. Do not tighten them too much.

Figure 23 - Fixation of the gradient coil



25. Remove the guiding strips from the magnet bore (position 16 and 22) and mount back the shimrails.
26. Measure the isolation between the Hybrid box and the gradient coil RF screen (earth point on front top).

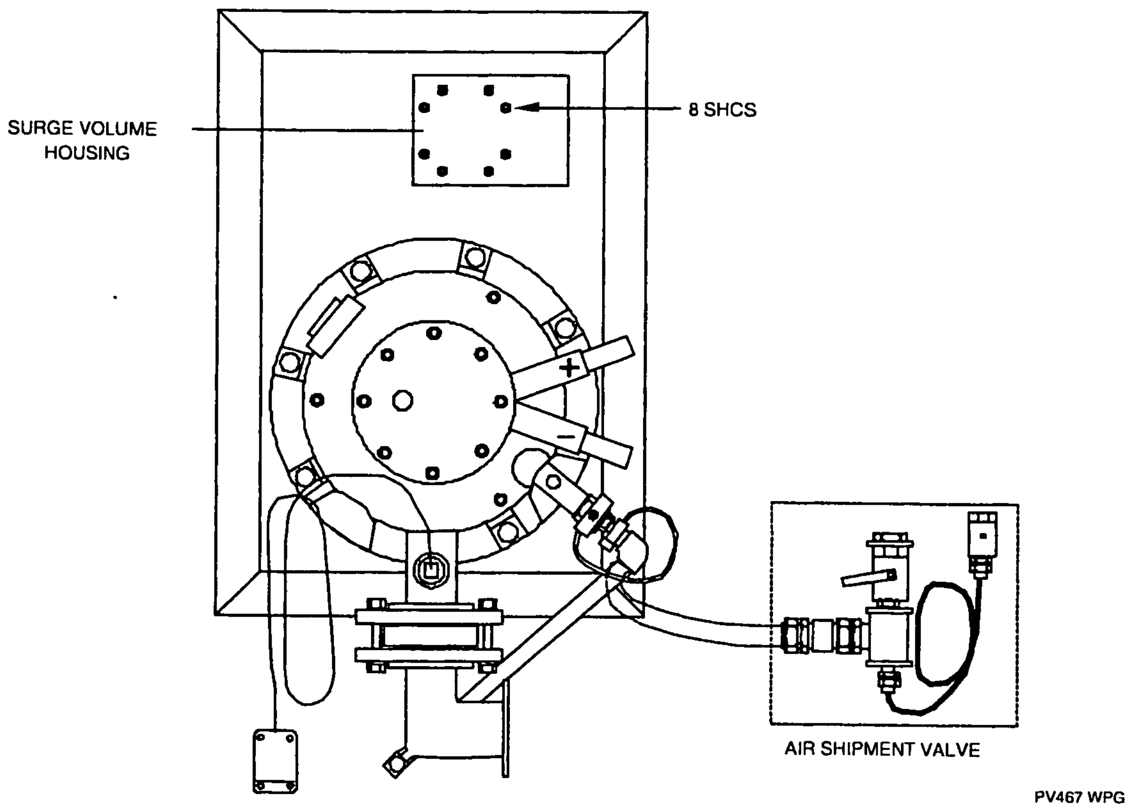
10. REPLACE THE AIR SHIPMENT VALVE

To prevent air from entering the magnet during air transport, an air shipment valve has been mounted to the
To prevent air from entering the magnet during air transport, an air shipment valve has been mounted to the
turret.

Procedure for the helium exhaust (see Figure 24):

1. Release the magnet pressure by opening the valve on the air shipment valve.
2. Disconnect the air pressure release valve.
3. Mount the flexible helium exhaust hose to the fixed helium exhaust valve assembly.
4. Close the ball valve (yellow valve).
5. Return the air shipment valve to Best.

Figure 24 - Helium exhaust type 4 with air shipment valve



MOUNTING THE GAS EXHAUST LINES A gas exhaust line for inside the RF enclosure is supplied with the system. The RF feedtrough is supplied together with the pre-delivery kit to be mounted by the RF cage supplier. The external exhaust pipes should also be present.

Mount the remaining parts of the gas exhaust line according to the following procedure.
Refer to drawing Z3-4.5 of the drawings manual

CAUTION

Take care that the gas exhaust line does not touch other conductors in its neighborhood (e.g. false ceiling). This may cause the so-called 'spike' artifact.

WARNING

Never cut the flexible exhaust pipe in order to adapt its length. In that case the connection between the flexible pipe and its flange is too weak so that you remain with an unsafe situation. The glue that is present between pipe and flange is essential for its strength.

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1. Mount the flexible hose (500-mm) to the RF feedtrough interface (detail B of drawing Z3-4.5).
2. Mount the other flexible hose (1500-mm) to the exhaust of the magnet (detail D of drawing Z3-4.5).
3. Mount the hooks to the ceiling by screwing them onto the M5 bolts prepared by the RF supplier (detail F). The distance between two hooks should be between 500 and 800 mm).
4. Mount the pipes between the two flexible hoses. Depending of the distance use the 1000-mm pipes and/or the 500-mm pipe. To mount the flexible hose to a pipe see detail E. To mount two pipes together see detail C. Mount the pipes to the ceiling (detail A-A and detail F of drawing Z3-4.5).
5. Mount the drip-tray to the gas exhaust line with the strings (detail A-A of drawing Z3-4.5). The drip-tray may not touch the RF feedtrough and must be mounted so that water in this gutter will run to the magnet. (In case of a quench, the liquid air is caught in the drip-tray and will drip on the magnet outer housing, where it evaporates).
6. Fasten all bolts.

11. CLOSING THE MAGNET TRANSPORT PASSAGE

The transport opening may be closed when all large and heavy items have been brought into the examination room. As the passage forms part of the shielded RF room (examination room), proceed with the utmost care. The cage builders instructions have to be followed to ensure a properly shielded room.